

Solve equivalence problems

Use the scaffold, show each step and ask for an adult check when marked.

Name: _____ Date: _____

1. Read the model: Complete: $9/25 = \underline{\quad} = \underline{\quad}\%$. Copy or complete the first step.

2. Write $7/20$ as a decimal.

3. Write 0.625 as a percentage.

4. Miro says: Miro moves the decimal point without considering the value. Is this correct? Explain.

Teacher answers and checking prompts

1. Read the model: Complete: $9/25 = \underline{\quad} = \underline{\quad}\%$. Copy or complete the first step.
Answer 0.36 and 36%.
2. Write $7/20$ as a decimal.
Answer 0.35.
3. Write 0.625 as a percentage.
Answer 62.5%.
4. Miro says: Miro moves the decimal point without considering the value. Is this correct? Explain.
Answer Use a hundred-square model or multiplication and division by 100, then check against a known benchmark.

Solve equivalence problems

Work independently. Show a complete method and check at least one answer.

Name: _____ Date: _____

1. Complete: $9/25 = \underline{\quad} = \underline{\quad}\%$.

6. Check this result using an inverse, estimate or known fact: Write 0.625 as a percentage.

2. Write $7/20$ as a decimal.

7. Prove or explain your reasoning: Write 35% as a fraction in simplest form.

3. Write 0.625 as a percentage.

8. Identify and correct this misconception: Miro moves the decimal point without considering the value.

4. Solve this independently and show every stage: Complete: $9/25 = \underline{\quad} = \underline{\quad}\%$.

9. Write a complete sentence explaining how you checked one answer.

5. Use a second method or representation for: Write $7/20$ as a decimal.

10. Write and solve a similar question to: Write 0.625 as a percentage.

Teacher answers and checking prompts

1. Complete: $9/25 = \underline{\quad} = \underline{\quad}\%$.
Answer 0.36 and 36%.
2. Write $7/20$ as a decimal.
Answer 0.35.
3. Write 0.625 as a percentage.
Answer 62.5%.
4. Solve this independently and show every stage: Complete: $9/25 = \underline{\quad} = \underline{\quad}\%$.
Answer 0.36 and 36%.
5. Use a second method or representation for: Write $7/20$ as a decimal.
Answer Expected result: 0.35. Method or representation may vary.
6. Check this result using an inverse, estimate or known fact: Write 0.625 as a percentage.
Answer Expected result: 62.5%. A valid check must be shown.
7. Prove or explain your reasoning: Write 35% as a fraction in simplest form.
Answer $7/20$. The explanation must show why the method works.
8. Identify and correct this misconception: Miro moves the decimal point without considering the value.
Answer Use a hundred-square model or multiplication and division by 100, then check against a known benchmark.
9. Write a complete sentence explaining how you checked one answer.
Answer Answer should name an inverse, estimate, boundary, known fact or equivalent representation.
10. Write and solve a similar question to: Write 0.625 as a percentage.
Answer Answers vary. The new question must use the same mathematical structure and include a correct solution.

Solve equivalence problems

Prove, compare or generalise. Write enough reasoning for another pupil to follow.

Name: _____ Date: _____

1. Prove or explain your reasoning: Write 35% as a fraction in simplest form.

6. Create a new example that tests the same idea as: Write 0.625 as a percentage.

2. Prove the answer to this question, rather than only calculating it: Complete: $9/25 = \underline{\quad} = \underline{\quad}\%$.

7. Find a second method or representation. Compare its efficiency with your first method.

3. Solve this in two different ways and compare them: Write $7/20$ as a decimal.

8. Write one always, sometimes or never statement about today's learning and justify it.

4. Work backwards from the answer to reconstruct the question: 62.5%.

9. Create a believable incorrect solution. Identify the exact step where the reasoning breaks.

5. Explain why this reasoning fails: Miro moves the decimal point without considering the value.

10. Change one value or condition in a question. Explain what changes in the solution and what stays the same.

Teacher answers and checking prompts

1. Prove or explain your reasoning: Write 35% as a fraction in simplest form.
Answer 7/20. The explanation must show why the method works.
2. Prove the answer to this question, rather than only calculating it:
Complete: $9/25 = \frac{\quad}{\quad} = \frac{\quad}{\quad}\%$.
Answer Expected result: 0.36 and 36%. Proof or justification must match the lesson structure.
3. Solve this in two different ways and compare them: Write 7/20 as a decimal.
Answer Expected result: 0.35. Two valid methods and a comparison are required.
4. Work backwards from the answer to reconstruct the question: 62.5%.
Answer One valid reconstruction is: Write 0.625 as a percentage. Other equivalent questions are acceptable.
5. Explain why this reasoning fails: Miro moves the decimal point without considering the value.
Answer Use a hundred-square model or multiplication and division by 100, then check against a known benchmark.
6. Create a new example that tests the same idea as: Write 0.625 as a percentage.
Answer Answers vary. The example and solution must preserve the mathematical structure.
7. Find a second method or representation. Compare its efficiency with your first method.
Answer Answers vary. Comparison should refer to accuracy, number structure and clarity.
8. Write one always, sometimes or never statement about today's learning and justify it.
Answer Answers vary. A valid example and counterexample should support the classification.
9. Create a believable incorrect solution. Identify the exact step where the reasoning breaks.
Answer Answers vary. The error must be mathematically relevant and the correction must be precise.
10. Change one value or condition in a question. Explain what changes in the solution and what stays the same.
Answer Answers vary. The explanation must distinguish the mathematical structure from the changed value.